

浙江大学 2018 - 2019 学年夏学期

《动物生物统计学》课程期末考试试卷

课程号: 17120130, 开课学院: 动物生物统计学

考试试卷: ☒ A 卷、B 卷 (请在选定项上打 $\checkmark$)

考试形式: ☒ 闭、开卷 (请在选定项上打 $\checkmark$), 允许带 计算器 入场

考试日期: 2019 年 6 月 27 日, 考试时间: 120 分钟

诚信考试, 沉着应考, 杜绝违纪。

考生姓名: _____ 学号: _____ 所属院系: _____

题序	一	二	三	四	五	六	七	八	总分
得分									
评卷人									

Table 1. Reference

$F_{0.975,35,25} = 0.488$	$F_{0.025,35,25} = 2.146$	$F_{0.05,35,25} = 1.892$	$F_{0.95,35,25} = 0.548$
$F_{0.975,36,26} = 0.494$	$F_{0.025,36,26} = 2.114$	$F_{0.05,36,26} = 1.869$	$F_{0.95,36,26} = 0.554$
$t_{0.05,60} = 1.671$	$t_{0.025,60} = 2$	$t_{0.05,62} = 1.67$	$t_{0.025,62} = 1.999$
$t_{0.05,1} = 6.313$	$t_{0.025,1} = 12.706$	$\chi^2_{0.05,2} = 5.99$	$\chi^2_{0.025,3} = 7.38$
$\chi^2_{0.01,2} = 9.21$	$\chi^2_{0.05,3} = 7.81$	$\chi^2_{0.025,3} = 9.35$	$\chi^2_{0.01,3} = 11.34$

一、Term Explanation (Each 3 score, total 15 score)

1. Geometric mean
2. Percentile
3. Accompany probability
4. Population
5. Sample correlation coefficient



二、Short Answer Questions (Each 5 score, total 25 score)

1. Write down the linear model of one-way ANCOVA. Describe the basic assumptions for one-way ANCOVA.
2. Prove sampling variance of sample variance equals $\frac{2\sigma^4}{n-1}$.
3. For the set of numbers: 1, 3, 7, 9, 11, 19, 20, calculate the 1st Quartile and median.
4. For multiple comparison in ANOVA, what is the difference between LSD and Bonferroni's methods.
5. For a regression model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$, we have obtained n groups of data $(Y_1, \dots, Y_n)$. Then, what are the degree of freedoms for SST, SSR, and SSE?

三、Calculation (Each 15 score, total 60 score)

1. To study the effect of gender on the amount of red blood cell (RBC, unit: 10000/mm³) in adults, 36 males and 26 females were randomly chosen for the test. Then, the averages of RBC in males and females were obtained as 465.13 and 422.16, respectively, with sample variances of 54.80² and 49.30², respectively. Based on the information above, can we conclude that there is an effect of gender on the amount of RBC in adults? Explain why. (The amount of RBC is normally distributed. $\alpha = 0.05$)

2. Early human geneticist liked to study the relationship of some traits between generations. In table 2, we have body height values collected from 12 pairs of fathers and sons. Use X denotes body height of fathers', and Y does those of sons'.
 - a) Establish a regression model using Y as dependent variable.
 - b) Calculate the correlation coefficient of X and Y.
 - c) If we want to test whether the correlation coefficient is zero or not, T-test would be used. In this case, write down the T-test formula and explain what the degree of freedom is.

Table 2 body height (in inches)

X	65	63	67	64	68	62	70	66	68	67	69	71
Y	68	66	68	65	69	66	68	65	71	67	68	70



3. The Mendelian Law of Segregation tells us that expected ratio of offspring genotype would be 1:2:1, when the mating diploid parents' genotypes are Aa and Aa. Now we collected 200 piglets, which all are the offspring of parents with Aa genotype. After genotyping, there are 60 AA, 110 Aa, and 30 aa in the piglets. Therefore, does the inheritance of A and a alleles really follows Mendelian Law of Inheritance? ($\alpha = 0.05$)

4. A pet company tested three formulations of shampoo for the clearance of stains on pet fur. For the experiment, 27 pets were selected and 9 were randomly assigned to one of the three shampoo formulations. If use 1 to 10 to describe the outcome of the test with 1 meaning completely messy and 10 meaning perfectly clean. Then, the data are summarized as follows, $\bar{X}_{..} = 5.11$, $\bar{X}_{1.} = 3.67$, $\bar{X}_{2.} = 5.78$, $\bar{X}_{3.} = 5.89$. Fill the ANOVA table ($\alpha = 0.05$)

ANOVA table

Source of Variance	SS	MS	df	F	Critical F value
Shampoo					$F_{(), (), ()}$
Error	28.44				
Total					

四、Extra problem (Extra 10 score will be awarded)

For a random variable X_{ijk} , three factors are suggested to have impacts on it. These factors are A, B, C. Now we want to design a complete randomized design experiment to test which one or ones really play a role in influencing the random variable. For factor A, there are I levels. For factor B, there are J levels. For factor C, there are K levels. Within each treatment, there are L repeats.

- Write down the linear model for the design.
- Write down the equations, using which we can calculate the SST, SSE, and SS (main/interaction effects).
- Write down the degree of freedom for respective SS.

17	11	71	20	30	07	50	80	00	70	80	70	X
07	20	70	17	70	30	80	50	70	80	80	90	Y



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题序	一	二	三	四	五	六	七	八	总分
得分									
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Table 1 Reference

$F_{0.025,9,9} = 4.026$	$F_{0.975,9,9} = 0.248$	$F_{0.005,9,9} = 6.541$	$F_{0.995,9,9} = 0.153$
$F_{0.05,9,9} = 3.179$	$F_{0.95,9,9} = 0.314$	$t_{0.025,12} = 2.179$	$t_{0.025,10} = 2.228$
$t_{0.025,13} = 2.160$	$t_{0.05,10} = 1.812$	$t_{0.05,12} = 1.782$	$t_{0.05,13} = 1.771$
$\chi^2_{0.05,2} = 5.99$	$\chi^2_{0.05,1} = 3.84$	$\chi^2_{0.025,2} = 7.38$	$\chi^2_{0.025,1} = 5.02$

一、Terms Explanation. (Each 3 score, total 15 score)

1. Biostatistics
2. Degree of freedom
3. Interaction effect
4. SSB (between) in one-way ANOVA
5. Latin square design



二、Short Answer Questions. (Each 5 score, total 25 score)

1. Write down the linear model for two-way ANOVA with repeats. Describe the basic assumptions for it.
2. What will happen for a χ^2 distribution ($df=n$), when $n \rightarrow +\infty$. Then what is the expectation and variance of the new distribution?
3. Describe what are type I and II errors? How are confidence level and statistical power expressed by the two errors?
4. For homogeneity of variance test in one-way ANOVA, which one can we use for unbalanced design, Bartlett, Hartley F or Cochran?
5. If we know the relationship of Y and X that satisfies $Y = aX^b\epsilon$, how can we do proper transformation to do regression analysis?

三、Calculation (Each 15 score, total 60 score)

1. We want to compare whether there is weight difference between pig breed A and B at certain age. Then 10 pigs from each breed were randomly chosen, and the data were calculated as follows: $\bar{x}_A = 30.97$, $S_A = 26.7$, $\bar{x}_B = 21.79$, $S_B = 12.1$. From these data, what kind of conclude can we make regarding the weight difference? (Normality is assumed. $\alpha = 0.05$)
2. Consider a mouse model of autosomal recessive disease with aa genotype denoting disease and AA/Aa being normal. A pair of Aa mice produced 20 pups, with 13 normal and 7 diseased. Thus, does the model follow a recessive inheritance? ($\alpha = 0.05$)
3. The table 2 lists a BCA method for measurement of protein concentration.
 - a) Calculate what are the concentrations of samples when A_{562} are 0.43 and 0.70
 - b) What does it mean, when $\frac{MSR}{MSE} > F_{0.05,1,5}$?

Table 2

Pr (mg/ml)	0.025	0.050	0.10	0.20	0.30	0.40	0.50
A_{562}	0	0.06	0.13	0.22	0.35	0.45	0.66

4. A pet company tested three formulations of shampoo for the clearance of stains on pet fur. For the experiment, 27 pets were selected and 9 were randomly assigned to



one of the three shampoo formulations. If use 1 to 10 to describe the outcome of the test with 1 meaning completely messy and 10 meaning perfectly clean. Then, the data are summarized as follows, $\bar{X}_{..} = 5.11$, $\bar{X}_{1.} = 3.67$, $\bar{X}_{2.} = 5.78$, $\bar{X}_{3.} = 5.89$. Fill the ANOVA table ($\alpha = 0.05$)

ANOVA table

Source of Variance	SS	MS	df	F	Critical F value
Shampoo					$F_{(),(),()}$
Error	28.4				
Total					

四、Extra problem (Extra 10 score will be added)

Explain in a brief and clear way why when $\frac{MSA}{MSE} > F_{\alpha, df_A, df_E}$, the null hypothesis will be rejected in balanced one-way fixed-effect ANOVA.

